

# HDOC DAY-24

**Title:** Increasing Pattern Generation using Parameterless Functions and Nested Loops

## 1. Problem Statement

Write a menu-driven C program using a switch statement to print various "Increasing patterns" selected by the user. Use nested loops without arrays. Each pattern must be implemented in a separate parameterless function returning void (void function\_name(void)):

1. **Increasing right-angled triangle:** Print asterisks with  $n$  rows.
2. **Increasing row-number triangle:** Print row number repeated  $i$  times in row  $i$ .
3. **Increasing natural-number pattern:** Print natural numbers from 1 up to  $i$  in row  $i$ .
4. **Consecutive-number triangle:** Print continuous consecutive natural numbers across rows.
5. **Increasing alphabet pattern:** Print uppercase letters from 'A' up to the letter corresponding to row length ( $1 \leq n \leq 26$ ).
6. **Exit:** Terminate program loop.

## 2. Algorithms

### Main Control Flow

1. Display Increasing Patterns Menu inside a do-while loop.
2. Read choice and route through switch statement to execute corresponding parameterless function.
3. Exit when choice 6 is entered.

### Pattern Functions

- **print\_right\_angled\_triangle():** Read  $n$ . Loop  $i$  from 1 to  $n$ , inner loop  $j$  from 1 to  $i$ : print \*.
- **print\_increasing\_row\_number():** Read  $n$ . Loop  $i$  from 1 to  $n$ , inner loop  $j$  from 1 to  $i$ : print  $i$ .
- **print\_increasing\_natural\_number():** Read  $n$ . Loop  $i$  from 1 to  $n$ , inner loop  $j$  from 1 to  $i$ : print  $j$ .
- **print\_consecutive\_number():** Initialize count = 1. Read  $n$ . Loop  $i$  from 1 to  $n$ , inner loop  $j$  from 1 to  $i$ : print count++.
- **print\_increasing\_alphabet():** Read  $n$ . Loop  $i$  from 1 to  $n$ , inner loop  $j$  from 0 to  $i - 1$ : print 'A' +  $j$ .

## 3. Test Case Table

| <b>Tes<br/>t<br/>Cas<br/>e</b> | <b>Pattern /<br/>Option</b>   | <b>Input<br/>n</b> | <b>Expecte<br/>d<br/>Pattern<br/>Output</b> | <b>Actual<br/>Output</b>              | <b>Stat<br/>us</b> |
|--------------------------------|-------------------------------|--------------------|---------------------------------------------|---------------------------------------|--------------------|
| <b>1</b>                       | Right-<br>Angled<br>Triangle  | $n = 4$            | * rows:<br>1, 2, 3, 4<br>stars              | Match<br>ed<br>expect<br>ed<br>output | Pass               |
| <b>2</b>                       | Row-<br>Number<br>Triangle    | $n = 4$            | Rows: 1<br>/ 2 2 / 3<br>3 3 / 4 4<br>4 4    | Match<br>ed<br>expect<br>ed<br>output | Pass               |
| <b>3</b>                       | Natural-<br>Number<br>Pattern | $n = 4$            | Rows: 1<br>/ 1 2 / 1<br>2 3 / 1 2<br>3 4    | Match<br>ed<br>expect<br>ed<br>output | Pass               |
| <b>4</b>                       | Consecuti<br>ve-              | $n = 4$            | Rows: 1<br>/ 2 3 / 4                        | Match<br>ed<br>expect                 | Pass               |

| Test Case | Pattern / Option       | Input n    | Expected Pattern Output                     | Actual Output                         | Status |
|-----------|------------------------|------------|---------------------------------------------|---------------------------------------|--------|
|           | Number Triangle        |            | 5 6 / 7 8<br>9 10                           | ed<br>output                          |        |
| 5         | Alphabet Pattern       | $n = 4$    | Rows: A<br>/ A B / A<br>B C / A B<br>C D    | Match<br>ed<br>expect<br>ed<br>output | Pass   |
| 6         | Invalid Alphabet Range | $n = 30$   | Error<br>messag<br>e ( $1 \leq n \leq 26$ ) | Error<br>messa<br>ge<br>shown         | Pass   |
| 7         | Exit Option            | Choice = 6 | "Exiting<br>program<br>..."                 | Progra<br>m<br>exited                 | Pass   |

#### 4. Source Code Listing

```
#include <stdio.h>

void print_right_angled_triangle(void) {
    int n;
    printf("\n--- Increasing Right-Angled Triangle ---\n");
```

```
printf("Enter number of rows (n): ");
scanf("%d", &n);
if (n <= 0) { printf("Error: Input must be a positive integer.\n"); return; }
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) printf("*");
    printf("\n");
}
}
```

```
void print_increasing_row_number(void) {
    int n;
    printf("\n--- Increasing Row-Number Triangle ---\n");
    printf("Enter number of rows (n): ");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Input must be a positive integer.\n"); return; }
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) printf("%d ", i);
        printf("\n");
    }
}
```

```
void print_increasing_natural_number(void) {
    int n;
    printf("\n--- Increasing Natural-Number Pattern ---\n");
    printf("Enter number of rows (n): ");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Input must be a positive integer.\n"); return; }
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) printf("%d ", j);
        printf("\n");
    }
}
```

```
void print_consecutive_number(void) {
    int n;
    printf("\n--- Consecutive-Number Triangle ---\n");
    printf("Enter number of rows (n): ");
    scanf("%d", &n);
    if (n <= 0) { printf("Error: Input must be a positive integer.\n"); return; }
    int count = 1;
```

```

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) printf("%d ", count++);
        printf("\n");
    }
}

void print_increasing_alphabet(void) {
    int n;
    printf("\n--- Increasing Alphabet Pattern ---\n");
    printf("Enter number of rows (1 <= n <= 26): ");
    scanf("%d", &n);
    if (n < 1 || n > 26) { printf("Error: n must be between 1 and 26.\n"); return;
}
    for (int i = 1; i <= n; i++) {
        for (int j = 0; j < i; j++) printf("%c ", 'A' + j);
        printf("\n");
    }
}

int main(void) {
    int choice;
    do {
        printf("\n===== INCREASING PATTERNS MENU\n");
        printf("1. Increasing right-angled triangle\n");
        printf("2. Increasing row-number triangle\n");
        printf("3. Increasing natural-number pattern\n");
        printf("4. Consecutive-number triangle\n");
        printf("5. Increasing alphabet pattern\n");
        printf("6. Exit\n");
        printf("Enter choice (1-6): ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: print_right_angled_triangle(); break;
            case 2: print_increasing_row_number(); break;
            case 3: print_increasing_natural_number(); break;
            case 4: print_consecutive_number(); break;
            case 5: print_increasing_alphabet(); break;
            case 6: printf("Exiting program...\n"); break;
        }
    } while (choice != 6);
}

```

```
        default: printf("Error: Invalid choice!\n"); break;
    }
} while (choice != 6);
return 0;
}
```

#### 4. Execution & Verification Screenshots

```
Session Actions Edit View Help

(kali@kali)-[~]
$ mkdir HDOC_Day24
cd HDOC_Day24
(kali@kali)-[~/HDOC_Day24]
$ nano day24.c
(kali@kali)-[~/HDOC_Day24]
$ gcc day24.c -o day24
(kali@kali)-[~/HDOC_Day24]
$ ./day24

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter choice (1-6): 1

--- Increasing Right-Angled Triangle ---
Enter number of rows (n): 4
*
**
***
****
      Triangle

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter choice (1-6): 2

--- Increasing Row-Number Triangle ---
Enter number of rows (n): 4
1
2 2
3 3 3
4 4 4 4

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter choice (1-6): 3

--- Increasing Natural-Number Pattern ---
Enter number of rows (n): 4
1
1 2
1 2 3
1 2 3 4

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter choice (1-6): 4

--- Consecutive-Number Triangle (Floyd's Triangle) ---
Enter number of rows (n): 4
1
2 3
4 5 6
7 8 9 10

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter choice (1-6): 5

--- Increasing Alphabet Pattern ---
Enter number of rows (1 ≤ n ≤ 26): 4
A
A B
A B C
A B C D

===== INCREASING PATTERNS MENU =====
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
6. Exit
Enter choice (1-6): 6
Exiting program...

(kali@kali)-[~/HDOC_Day24]
```